PIEAS

Pakistan Institute of Engineering and Applied Sciences

Department of Electrical Engineering

Designing of a 12V 0.5A DC Power Supply

Complex Engineering Problem

EE-120

Electronic Devices and Circuits

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Date Of Submission:

June 1, 2023

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1.1 Abstract:

The purpose of this report is to present the design and analysis of a DC power supply with a regulated output voltage of 12V and a minimum current output of 0.5A. The power supply will be used to provide a stable and reliable source of DC power for various electronic devices and applications. The report outlines the design specifications, the components used, the circuit diagram, and the analysis of the power supply's performance. To achieve a regulated output voltage of 12V, the power supply utilizes voltage regulation techniques such as feedback control mechanisms or voltage regulation circuits. This ensures that the output voltage remains constant despite variations in input voltage or changes in load current. Additionally, current-limiting mechanisms are implemented to restrict the output current to a safe level, preventing potential damage to the power supply or connected devices. The components utilized in the power supply design include a transformer, rectifier circuit, filter circuit, and voltage regulator. The transformer steps down the AC mains voltage to a lower AC voltage, while the rectifier circuit converts AC voltage to DC voltage. The filter circuit reduces ripple voltage, and the voltage regulator maintains a constant output voltage of 12V. The specific values and specifications for these components would depend on the chosen design and desired performance characteristics. The circuit diagram illustrates the connections and arrangement of the components, providing a visual representation of how the power supply is constructed. This diagram serves as a reference for the analysis of the power supply's performance. The performance of the power supply is analyzed by measuring the output voltage, current, and ripple voltage. These measurements are compared against the design specifications to assess the power supply's adherence to the desired parameters. Any deviations or limitations in performance are discussed, along with potential improvements. Overall, the design and analysis of the DC power supply with a regulated output voltage of 12V and a minimum current output of 0.5A aim to provide a stable, reliable, and efficient source of DC power for a range of electronic devices and applications. The report serves as a comprehensive guide for understanding the design process, components used, circuitry, and performance characteristics of the power supply.

1.2 Introduction:

In today's electronic devices, a stable and reliable power supply is essential. DC power supplies play a crucial role in providing a constant source of power for various applications. This report focuses on the design and analysis of a DC power supply with an output voltage of 12V and a maximum current output of 0.5A.

A 12-volt, 0.5-ampere DC power supply provides a stable and regulated output voltage of 12 volts and a maximum output current of 0.5 amperes. The theoretical background behind this type of power supply involves several key concepts:

Voltage Regulation: The power supply is designed to maintain a constant output voltage of 12 volts, regardless of variations in input voltage or changes in load current. This is typically achieved through the use of voltage regulation techniques, such as feedback control mechanisms or voltage regulation circuits.

Current Limiting: The power supply is capable of providing a maximum output current of 0.5 amperes. This means that it can deliver up to 0.5 amperes of current to the connected load while maintaining the desired voltage level. To prevent excessive current flow that could potentially damage the power supply or the load, current-limiting mechanisms may be implemented to restrict the output current to a safe level.

Power Calculation: The power output of the power supply can be calculated by multiplying the output voltage (12 volts) by the output current (0.5 amperes). In this case, the power output would be 6 watts (12V x 0.5A = 6W).

Efficiency: The efficiency of the power supply refers to how effectively it converts input power into usable output power. The efficiency is typically expressed as a percentage and can be calculated by dividing the output power by the input power and multiplying by 100. However, the specific efficiency of the power supply would depend on the design and components used.

Safety Considerations: It is important to ensure that the power supply meets safety standards and regulations to protect against electrical hazards. This may involve features such as short-circuit protection, overcurrent protection, overvoltage protection, and insulation to ensure safe and reliable operation.

Overall, the theoretical background of a 12-volt, 0.5-ampere DC power supply involves concepts such as voltage regulation, current limiting, power calculation, efficiency, and safety considerations to provide a stable and regulated power output to the connected load.

1.3Design Specifications:

The design specifications for the DC power supply are as follows:
Output Voltage: 12V (regulated)
Maximum Current Output: 0.5A
Input Voltage: AC mains voltage (e.g., 110V or 220V)

Efficiency: Greater than 80%
Ripple Voltage: Less than 50mV

1.4 Objective:

The purpose of this lab was to design a regulated 12V DC power supply from a 220-230 V AC source.

1.4 Equipment required:

The following components are used in the design of the DC power supply:

component	Model number	Voltage (v)	Current (Amp)	Cost (Rs)
Transformer	Local	230V-12V	2A	500Pkr
Bride rectifier	Local		3A	50Pkr
capacitor	1500uF	50V		40Pkr
Voltage regulator	LM 7812 CV	12V	1.5A	50Pkr
resistor	22Ω 10W			20Pkr

Table 1: Equipment Information

1.4.1Transformer:

Steps down the AC mains voltage to a lower AC voltage.

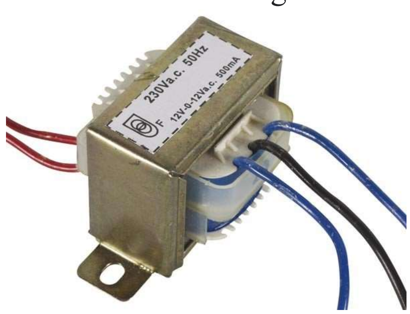

Figure 1: Transformer

Specifications:

- **Input Voltage: 230V AC**

This indicates the voltage required at the primary side of the transformer for proper operation. In this case, it is 230V AC.

- **Output Voltage: 12V AC**

This specifies the voltage that will be produced at the secondary side of the transformer. It is 12V AC in this case.

- **Power Rating: 24VA (Volt-Ampere) or 28.8W**

The power rating is calculated by multiplying the output voltage by the maximum current rating. In this case, it is 12V x 2A = 24VA or 28.8W.

- **Frequency: 50Hz or 60Hz**

Transformers are designed to operate at a specific frequency, either 50Hz (common in many countries) or 60Hz (common in North America and some other regions).

- **Turns Ratio:**

The turns ratio determines the voltage conversion ratio between the primary and secondary windings. For a 230V to 12V transformer, the turns ratio would typically be 19.17:1

1.4.2Rectifier Circuit:

Converts the AC voltage to DC voltage.

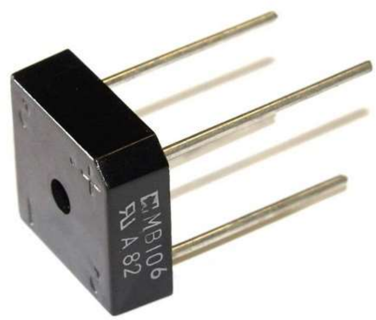

Figure 2: Bride Rectifier

Specifications:

The specifications of a bridge rectifier circuit typically depend on the specific diodes used and the intended application. However, here are some general specifications to consider:

- **Maximum Forward Voltage Drop (Vf):**

The forward voltage drop across each diode in the bridge rectifier circuit is an important specification. It represents the voltage loss when current flows through the diode. Typical values for silicon diodes used in bridge rectifiers range from 0.6 to 1.2 volts.

- **Maximum Reverse Voltage (Vr):**

The maximum reverse voltage refers to the maximum voltage that can be applied in the reverse direction across the diodes without causing breakdown. It is an important parameter to consider for the diodes used in the bridge rectifier. The value is typically specified as a peak or RMS voltage

1.4.3Filter Circuit:

Reduces the ripple voltage.

Information:

A filter circuit containing a 1 kΩ resistance and a 1500 µF capacitor in parallel can be used in a DC power supply for the following purposes:

1.4.4Smoothing Capacitor:

The capacitor in the filter circuit acts as a smoothing or reservoir capacitor. It helps to reduce the ripple voltage present in the output of the power supply. As the capacitor charges, it stores energy from the power supply during the peaks of the AC input voltage, and during the valleys, it releases the stored energy, providing a more stable and smoothed DC voltage.

- **Noise Filtering:**

The filter circuit helps to filter out high-frequency noise present in the DC power supply. Capacitors have the ability to block high-frequency components and allow lower-frequency signals to pass through. Therefore, the capacitor in the filter circuit can attenuate high-frequency noise, resulting in a cleaner DC output.

- **Voltage Stabilization:**

The filter circuit, in combination with the resistor, helps stabilize the output voltage by maintaining a constant voltage level. The resistor limits the charging and discharging current of the capacitor, allowing it to charge and discharge gradually. This helps to prevent sudden voltage variations and improves the stability of the output voltage.

- **Protection against Transients:**

The capacitor in the filter circuit can also provide protection against voltage transients or spikes. It can absorb and smooth out short-duration voltage spikes or transients that may occur in the power supply, thereby protecting downstream components from potential damage.

Overall, the filter circuit with the resistor and capacitor helps to improve the quality of the DC power supply by reducing ripple voltage, filtering noise, stabilizing the output voltage, and providing protection against transients. It is commonly used in power supplies for electronic devices where a stable and clean DC voltage is required for proper operation.

1.4.5Voltage Regulator:

Maintains a constant output voltage.

Figure 3: Voltage Regulator LM7812CV

LM7812

Electrical Characteristics (LM7812)

Refer to the test circuit, -40°C < T_J < 125°C, I_O = 500 mA, V_I = 19 V, C_I = 0.33 µF, C_O = 0.1 µF, unless otherwise specified.

Symbol	Parameter	Conditions		Min.	Typ.	Max.	Unit
V_O	Output Voltage	T_J = +25°C		11.5	12.0	12.5	V
		I_O = 5 mA to 1 A, $P_O \leq$ 15 W, V_I = 14.5 V to 27 V		11.4	12.0	12.6	
Regline	Line Regulation(12)	T_J = +25°C	V_I = 14.5 V to 30 V		10	240	mV
			V_I = 16 V to 22 V		3	120	
Regload	Load Regulation(12)	T_J = +25°C	I_O = 5 mA to 1.5 A		11	240	mV
			I_O = 250 mA to 750 mA		5	120	
I_Q	Quiescent Current	T_J = +25°C			5.1	8.0	mA
ΔI_Q	Quiescent Current Change	I_O = 5 mA to 1 A			0.1	0.5	mA
		V_I = 14.5 V to 30 V			0.5	1.0	
$\Delta V_O/\Delta T$	Output Voltage Drift(13)	I_O = 5 mA			-1		mV/°C
V_N	Output Noise Voltage	f = 10 Hz to 100 kHz, T_A = +25°C			76		µV
RR	Ripple Rejection(13)	f = 120 Hz, V_I = 15 V to 25 V		55	71		dB
V_{DROP}	Dropout Voltage	I_O = 1 A, T_J = +25°C			2		V
R_O	Output Resistance(13)	f = 1 kHz			18		mΩ
I_{SC}	Short-Circuit Current	V_I = 35 V, T_J = +25°C			230		mA
I_{PK}	Peak Current(13)	T_J = +25°C			2.2		A

Figure 4: Characteristic table of LM7812CV

Specifications:

The Lm7812CV is a commonly used voltage regulator integrated circuit (IC) from the 78xx series, specifically designed to provide a fixed output voltage of +12 volts. Here are the specifications of the Lm7812CV:

- **Input Voltage Range:**

The Lm7812CV requires an input voltage between 14 volts and 35 volts to regulate it down to a stable +12 volts.

- **Output Voltage:**

The output voltage of the Lm7812CV is fixed at +12 volts. It provides a stable DC voltage with a tolerance of ±4%.

- **Output Current:**

The maximum output current the Lm7812CV can handle is 1.5 Amperes. It is important to ensure that the load does not exceed this current limit to avoid overheating and potential damage to the IC.

- **Dropout Voltage:**

The dropout voltage of the LN7812CV is around 2 volts. This means that the input voltage must be at least 2 volts higher than the desired output voltage for the IC to regulate it properly.

- **Thermal Resistance:**

The thermal resistance of the LN7812CV is typically around 50°C/W. This value indicates how effectively the IC dissipates heat, and it is important to consider it when designing the heat sink or thermal management for the regulator.

1.5 Circuit Diagram:

A detailed circuit diagram for the DC power supply is provided, illustrating the connections and components used. The diagram includes the transformer, rectifier circuit, filter circuit, voltage regulator, and output capacitor. The values of the components are specified for the desired output voltage and current.

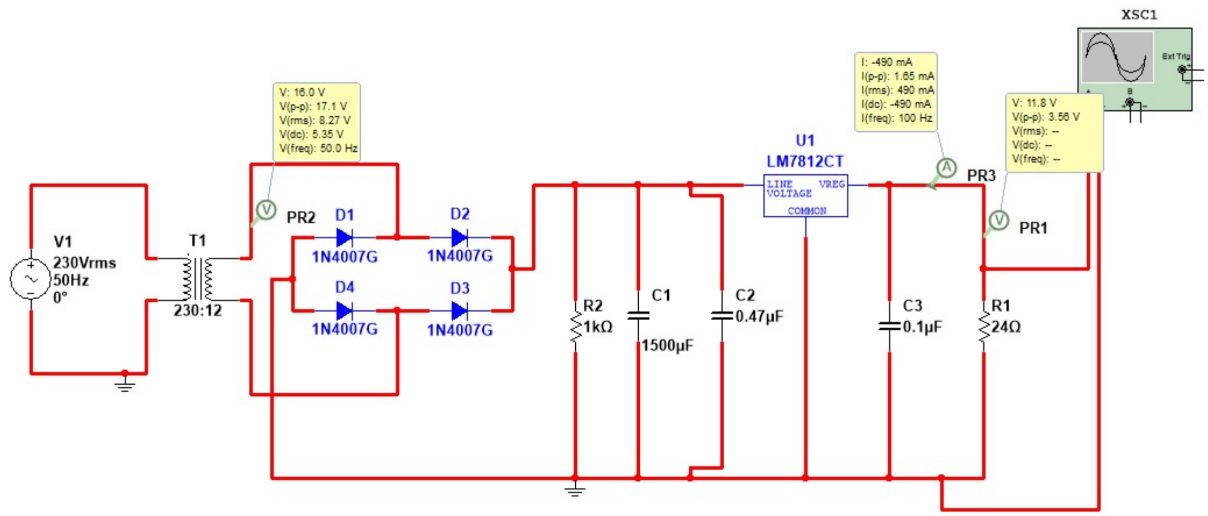

Figure 5: Simulated Circuit Diagram

Simulations:

Input signal

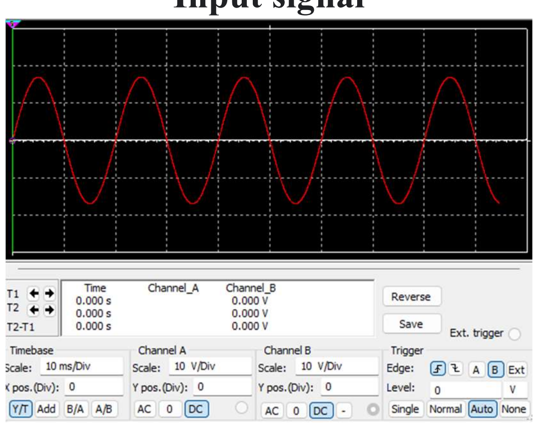

Figure 6: Simulated Input signal

Output signal

Figure 7: Simulated Output Circuit

Calculations

Regulator

Since we must design a 12 DC Supply, we will make use of LM 7812 Regulator. Dependingupon the input at the regulator we will proceed further with calculations:

From the Data sheet available for LM 7812, an input within the range (12-35)V is required to regulate a voltage of 12V, so taking:

$$V_{\text{reg in min}} = 12\ \text{V}$$

$$\text{Current Rating} = 0.5\ \text{A}$$

Diodes / Rectifier

Since the full wave rectifier involves a voltage drop of 0.7 + 0.7 V = 1.4 V and V reg in = 12 V,so the voltage required at the output of the transformer is:

$$V_{\text{sec min}} = 16 - 1.4\ \text{(Peak Voltage)}$$

$$V_{\text{sec min}} = 14.6\ \text{V (Peak Voltage)}$$

Here IN4001 diodes were utilized, having a current rating of 1A and PIV rating of 50 V which is acceptable.

Transformer

This voltage is given in rms whereas transformers are available in rms values, so converting to rms yields:

$$V_{\text{sec (rms) min}} = 0.707 * V_{\text{sec(peak)}}$$

$$V_{\text{sec (rms) min}} = 12.97\ \text{V}$$

Here we utilized the step-down transformer of turn Ration 19.33 having 16.97 V at secondary to have margin for loses throughout the circuit, so we get

$$V_{\text{sec (rms)}} = 12\text{V}$$

$$V_{\text{sec}}\ \text{Capacitor}$$

To select a capacitor, we need to incorporate ripple factor and the load resistance. So assuming the following for our circuit:

$$V_{\text{ripple(pp)}} = 3\%\ \text{of}\ V_{\text{reg out}} V_{\text{ripple(pp)}} = 0.03 * 9\text{V}\ V_{\text{ripple(pp)}} = 0.27\ \text{V}$$

$$\text{And} \quad R_L = 1\text{k}\Omega$$

So,

$$C = \frac{\text{I x 5}}{\text{V*2f}} * V_{\text{rec out}}$$

$$\text{C} = 1500 \text{ micro farad}$$

$$V_{(peak)} = 1.414 * 12 \text{ V} = 16.978 \text{ V} \approx 17 \text{ V}$$

Micro There is an option to decrease the load resistance to incorporate a higher capacitance. Here we fixed the load to be 1kΩ, and still increased the capacitance. This further minimized the ripple anderadicated any ups and downs associated with our rectified DC signal. Utilizing a capacitance of **1500µF** will yield a **ripple peak-to-peak voltage of 324mV** which is welcoming and acceptable.

Loading Circuit

In the loading circuit, resistance of 1kΩ was connected. The circuit wasbuilt to limit the voltage. Current of mA was flowing the external circuit. The current in the external circuit was around 2mA and was accommodated in loses throughout the circuit. Since the voltage drop across Resistor RL was around 5-7 V, an external capacitor of 10µF was connected in parallel to account for any ripple attenuation.

1.6 Practical Circuit

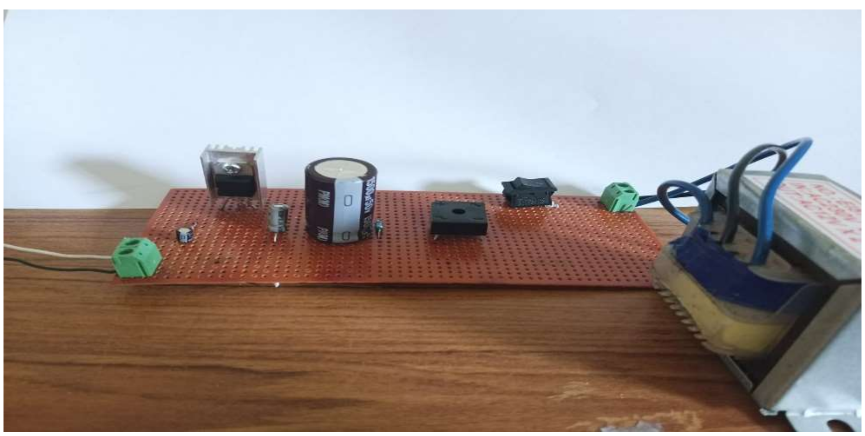

Figure 8: Practical Circuit

Waveform:

Input:

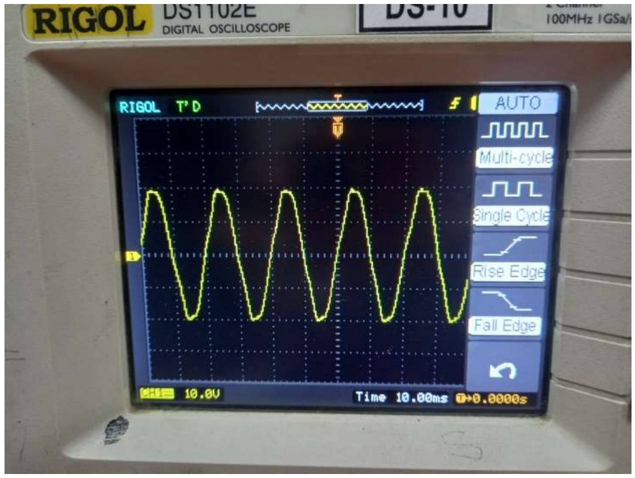

Figure 9: Experimental Input Signal

Output:

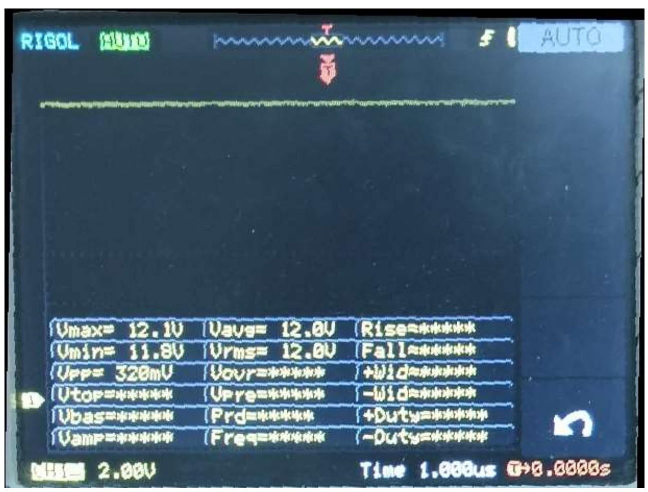

Figure 10: Experimental Output Signal

1.7 Discussion:

The performance of the DC power supply is analyzed based on the design specifications.
In the regulation of the output voltage of a 12V DC power supply, several measurements are commonly taken to ensure the voltage remains within the desired range. Here are some of the key measurements:

Output Voltage: The most important measurement is the actual output voltage of the power supply. This is typically measured using a voltmeter connected across the output terminals. In this case, it would be measured as 12V DC.

Load Regulation: Load regulation refers to how well the power supply maintains a stable output voltage as the load current changes. It is measured by varying the load current while monitoring the output voltage. Load regulation is typically expressed as a percentage and indicates the maximum deviation from the nominal voltage when the load changes.

Line Regulation: Line regulation measures the ability of the power supply to maintain a stable output voltage when the input voltage (line voltage) fluctuates. It is measured by varying the input voltage while monitoring the output voltage. Similar to load regulation, line regulation is expressed as a percentage and indicates the maximum deviation from the nominal voltage when the input voltage changes.

Ripple Voltage: Power supplies can introduce small fluctuations or ripples in the output voltage due to the conversion process. Ripple voltage refers to these fluctuations and is typically measured using an oscilloscope. It quantifies the amount of AC voltage superimposed on the DC output and is usually specified as peak-to-peak or RMS value.

Output Noise: Power supplies may also introduce electrical noise into the output voltage, which can affect the performance of sensitive devices. Output noise is measured using specialized equipment such as an oscilloscope or spectrum analyzer, and it is typically specified as a voltage or power spectral density. By monitoring and controlling these measurements, power supply regulators ensure that the output voltage remains stable and within the specified limits, providing reliable and consistent power to connected devices

Conclusion:

A DC Supply of -9V was developed using step down transformer, rectifier, capacitor, and regulator from 220 V AC. The DC output of the supply was verified by utilizing a loading circuit. The LED turned ON successfully and the voltage/current measures suggested that the rectification of the AC input voltage to DC was made possible using this circuit.